

Jibin Ou

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Objective: System Engineer, System Architect

EXPERIENCE

System Architect Sept. 2025–Current
Volvo Group, Gothenburg, Sweden

- Lead end-to-end system architecture for vehicle energy efficiency and guidance solutions across Adaptive and Classic AUTOSAR and Linux platforms;
- Define vehicle-level distribution architecture for digital map and geospatial services across platform consumers;
- Champion AI-driven engineering modernization, introducing agentic AI workflows to accelerate diagnostics and fault investigation;

System Architect 2022–2025
Volvo Cars, Gothenburg, Sweden

- Governed system architecture and platform integration for the Central Compute Platform across multi-disciplinary engineering domains;
- Directed service interface design and vehicle signal database integration across POSIX-based platform services and domain controller stacks;
- Established architecture standards for platform performance, end-to-end timing latency, and safety integrity;

System Engineer 2018–2022
Volvo Cars, Gothenburg, Sweden

- Drove vehicle system engineering for digital map and geolocation data integration across ADAS and automated driving systems;
- Led technical alignment and integration with external ecosystem partners (e.g., Google Android Automotive OS) and in-vehicle domain teams;

Software Developer Oct. 2016–2018
Honeywell, Shanghai, China

- Developed client applications and middleware for an enterprise IoT logistics platform; maintained Greater China cloud infrastructure;

Prior Experience (2011–2016)
Software Developer, Honeywell Movilizer, Mannheim, Germany 2015–2016
Research Assistant, ETH Zürich (HCI & Software Reliability Labs), Switzerland 2014
Development Intern, BMW Group (Vehicle Integration & Validation), Germany 2013
Research Assistant, Fraunhofer FIT (User-Centered Computing Group), Germany 2011–2012

EDUCATION

M.Sc., media informatics October 2010–April 2014
RWTH Aachen, Aachen, Germany
Visiting Student, computer science July 2013–April 2014
ETH Zürich, Zürich, Switzerland
B.Sc, information and computational science September 2006–June 2010
Sun Yat-sen University, Guangzhou, China

SELECTED PROJECTS

Agentic AI Workflow for Automated Issue & Fault Investigation Volvo Group

Intelligent diagnostic and analysis workflow for complex vehicle software defects

- Designed multi-agent LLM systems to autonomously parse vehicle issue reports, trace multi-repository source code, and analyze target system execution logs;
- Implemented multi-stage reasoning chains for hypothesis generation, root-cause evidence extraction, and automated regression test synthesis;
- Established closed-loop evaluation frameworks using historical failure cases and evidence validation gates to ensure rigor, explainability, and traceability;

Vehicle Energy Efficiency & Guidance System Architecture Volvo Group

End-to-end realization of vehicle energy management and driver guidance functions

- Architected full-stack vehicle energy management solution across adapter layers, dynamic configuration, real-time scoring, and user notifications;
- Established cross-environment test framework spanning Adaptive and Classic AUTOSAR and Linux platforms via SOME/IP services, utilizing Python;

Central Compute Platform Interface & Signal Architecture Volvo Cars

Platform architecture, signal database allocation, and service integration

- Managed end-to-end signal database allocation and service interfaces, transitioning legacy point-to-point signals into modular, service-oriented architectures (SOA);
- Optimized signal aggregation rates, serialization overhead, and thread scheduling to satisfy strict latency, throughput, and data integrity constraints;

Vehicle Map & Geolocation Information Subsystem Volvo Cars

Digital map and geospatial data distribution for active safety and driver assistance

- Designed onboard map service pipeline delivering road network geometry, curvature, and topological attributes to active safety applications;
- Standardized client query APIs, path reconstruction models, and vehicle caching strategies with Google Android Automotive OS and vehicle compute nodes;

SKILLS

Systems & Architecture: System Architecture, Automotive Systems, ADAS/AD, Central Compute Platform, Digital Map & Geospatial Services, SOME/IP, AUTOSAR (Adaptive & Classic), POSIX, Signal Databases (CAN/Ethernet), SOA
AI & Software: Agentic AI Workflows, LLM Prompt/Context Engineering, Python, C/C++, Java, Matlab, Linux, Git
Methods & Verification: Requirements Engineering, Automated Fault Investigation, CI/CD, LaTeX
Related coursework: Machine Learning, Human Computer Interaction

PUBLICATION

Jibin Ou, Martin Vechev, Otmar Hilliges. **An Interactive System for Data Structure Development**. In Proc. Computer Human Interaction (CHI) 2015, Seoul Korea. (average acceptance rate 20%)

LANGUAGE

English (professional proficiency), Swedish (advanced), German (advanced), Chinese (mother tongue), Cantonese (mother tongue)

HONORS & AWARDS

Won *Bravo Gold Recognition*, Honeywell, Shanghai, November 2016 and July 2017
Won *IDEA League Student Research Scholarship*, IDEA League, Zürich, 2013